

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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39. Vehicle Detection in a Video frame using OpenCV.

AIM:

Vehicle Detection in a Video frame.

PROGRAM:

```
import cv2

# Open the video
cap = cv2.VideoCapture(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\video2.mp4")

if not cap.isOpened():
    print("Error: Could not open video!")
    exit()

# Create background subtractor
fgbg = cv2.createBackgroundSubtractorMOG2(
    history=500,
    varThreshold=50,
    detectShadows=False
)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    # Apply background subtraction
    mask = fgbg.apply(frame)

    # Remove noise
    kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))
    mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)
    mask = cv2.dilate(mask, kernel, iterations=2)

    # Find contours
```

```

contours, _ = cv2.findContours(
    mask,
    cv2.RETR_EXTERNAL,
    cv2.CHAIN_APPROX_SIMPLE
)

for contour in contours:
    if cv2.contourArea(contour) > 1000:

        x, y, w, h = cv2.boundingRect(contour)

        cv2.rectangle(frame, (x, y), (x + w, y + h),
            (0, 255, 0), 2)

        cv2.putText(frame, "Vehicle",
            (x, y - 10),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.7,
            (0, 255, 0),
            2)

cv2.imshow("Vehicle Detection", frame)
cv2.imshow("Foreground Mask", mask)

if cv2.waitKey(30) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()

```

RESULT:

Thus the program is executed successfully.